

SIMATS ENGINEERING

CO2 - ASSESSMENT TOOL 2

Scenario-Based Questions

COURSE NAME: Deep Learning

COURSE CODE: MLA0408

COURSE OUTCOME COVERED

CO 2: Students will be able to apply supervised learning algorithms such as linear regression, classification models, decision trees, neural networks, support vector machines, and ensemble methods to solve real-world prediction and classification problems. They will gain the ability to select appropriate models, train them using datasets, and evaluate their performance. Students will also understand the strengths and limitations of different supervised learning approaches. (BTL-4)

CO Weightage: 15%

Assessment Tool Weightage: 40%

Duration: 90 minutes

Marks: 25

Code of Conduct:

I **M. Srilakshmi (192525404)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M. Srilakshmi

Scenario-Based Questions

CO-AT2

QUESTIONS: 05

Total Marks:25

1. Unsupervised Training of Neural Networks, Auto Encoders

A hospital wants to identify abnormal ECG patterns from thousands of unlabeled records. Which deep learning technique from the syllabus would you recommend? Justify your choice and explain the workflow.

Answer: Unsupervised Training of Neural Networks – Autoencoders

Recommended Deep Learning Technique:
Autoencoder (Unsupervised Learning)

Justification

A hospital has thousands of **unlabeled ECG (Electrocardiogram) records**, meaning there are no labels indicating whether an ECG is normal or abnormal. Since labeling medical data is time-consuming and expensive, an **Autoencoder** is the most suitable technique.

An autoencoder learns the normal patterns of ECG signals without requiring labeled data. After training on mostly normal ECGs, it can identify abnormal ECGs because they produce a **high reconstruction error**.

Workflow

1. Collect ECG Data

- Gather thousands of unlabeled ECG records.
- Perform preprocessing such as noise removal and normalization.

2. Train the Autoencoder

- Input ECG signals into the autoencoder.
- The encoder compresses the ECG into a lower-dimensional representation.
- The decoder reconstructs the original ECG signal.
- The network minimizes the difference between the original and reconstructed ECG.

3. Calculate Reconstruction Error

- Compute the difference between the original ECG and the reconstructed ECG.

4. Detect Abnormal Patterns

- **Low reconstruction error:** ECG is considered **normal**.
- **High reconstruction error:** ECG is considered **abnormal** because the model cannot accurately reconstruct unseen abnormal patterns.

5. Medical Review

- Flag abnormal ECGs for cardiologists to examine and diagnose.

Advantages

- Does not require labeled ECG data.
- Detects unknown or rare abnormalities.
- Reduces manual effort for doctors.
- Learns important features automatically.
- Suitable for large medical datasets.

Applications

- Arrhythmia detection
- Heart disease screening
- Anomaly detection in medical signals

- Patient health monitoring

Conclusion

An **Autoencoder** is the best choice because it is an **unsupervised deep learning technique** that learns the normal structure of ECG signals from unlabeled data. By measuring the reconstruction error, it effectively identifies abnormal ECG patterns, helping hospitals quickly detect potential heart diseases and support doctors in diagnosis.

2. Auto Encoders, Representation Learning

A retail company wants to reduce the dimensionality of customer purchase data before performing customer segmentation. Explain how an autoencoder can be used to solve this problem.

Answer: Autoencoders and Representation Learning for Customer Purchase Data

A retail company collects a large amount of customer purchase data, including product categories, purchase frequency, spending patterns, and transaction history. This data is often high-dimensional, making customer segmentation computationally expensive and less effective. An **autoencoder** can be used to reduce the dimensionality of this data while preserving the most important information.

How an Autoencoder Solves the Problem

1. Input Customer Data

- The original customer purchase data (high-dimensional features) is given as input to the autoencoder.

2. Encoder

- The encoder compresses the input into a smaller set of features called the **latent representation** or **bottleneck layer**.
- This compressed representation captures the essential purchasing patterns of customers.

3. Latent Representation

- The bottleneck layer contains fewer dimensions than the original data.
- It removes redundant and irrelevant information while retaining meaningful features.

4. Decoder

- The decoder reconstructs the original purchase data from the compressed representation.
- During training, the network minimizes the reconstruction error between the original and reconstructed data.

5. Customer Segmentation

- After training, only the encoder is used.
- The compressed representations are fed into clustering algorithms such as **K-Means** or **Hierarchical Clustering** to group customers with similar buying behaviors.

Advantages of Using Autoencoders

- Reduces data dimensionality without losing important information.
- Removes noise and redundant features.
- Improves clustering accuracy for customer segmentation.

- Reduces storage requirements and computational cost.
- Learns meaningful customer behavior patterns automatically.

Conclusion

Autoencoders provide an efficient way to perform **representation learning** by converting high-dimensional customer purchase data into a compact latent representation. This reduced-dimensional data can then be used for more accurate and efficient customer segmentation, helping the retail company identify customer groups and design personalized marketing strategies.

3. Width and Depth of Neural Networks, Activation Functions

A facial recognition system performs poorly when trained using a shallow neural network. Recommend architectural changes involving network depth and activation functions.

Answer: Width and Depth of Neural Networks, Activation Functions

A facial recognition system performs poorly when trained using a **shallow neural network** because it cannot learn the complex patterns and features present in facial images. Improving the network architecture by increasing its **depth** and selecting appropriate **activation functions** can significantly enhance performance.

Recommended Architectural Changes

1. Increase Network Depth

- Add more hidden layers to create a **deep neural network**.
- Deeper networks learn hierarchical features:
 - Early layers detect edges and textures.
 - Middle layers detect facial parts such as eyes, nose, and mouth.
 - Final layers recognize complete facial structures.

2. Adjust Network Width

- Increase the number of neurons in hidden layers if necessary.
- A wider network can learn more features, but excessive width may increase computation and overfitting.

3. Use ReLU Activation Function

- Replace sigmoid or tanh with **ReLU (Rectified Linear Unit)** in hidden layers.
- ReLU speeds up training, avoids the vanishing gradient problem, and improves convergence.

4. Use Softmax in the Output Layer

- For multi-person facial recognition, use the **Softmax** activation function.
- It outputs the probability of each class and selects the most likely identity.

Advantages

- Learns complex facial features more effectively.
- Improves recognition accuracy.
- Faster training with ReLU activation.
- Better handling of large image datasets.
- More robust feature extraction for different facial expressions and lighting conditions.

Conclusion

To improve facial recognition performance, the shallow neural network should be replaced with a **deeper neural network** having multiple hidden layers. Using **ReLU**

activation in hidden layers and **Softmax** in the output layer enables the model to learn complex facial features efficiently, resulting in higher recognition accuracy and better overall performance.

4. **Restricted Boltzmann Machines (RBMs), Unsupervised Training of Neural Networks**

A streaming service wants to recommend movies to users based on their viewing history without using labeled data. Explain how RBMs can be applied to build the recommendation engine.

Answer: Restricted Boltzmann Machines (RBMs) and Unsupervised Training of Neural Networks

A streaming service wants to recommend movies to users based on their viewing history without using labeled data. **Restricted Boltzmann Machines (RBMs)** are well-suited for this task because they learn hidden patterns from user–movie interactions through **unsupervised learning**.

How RBMs Build the Recommendation Engine

1. **Input User Viewing History**
 - The viewing history of users (watched/not watched or ratings) is used as the input data.
 - No labeled data is required.
2. **Visible Layer**
 - The visible layer represents users' movie preferences or viewing history.
3. **Hidden Layer**
 - The hidden layer learns hidden features such as movie genres, user interests, and viewing patterns.
 - These features are automatically discovered during training.
4. **Unsupervised Training**
 - The RBM is trained using algorithms such as **Contrastive Divergence (CD)**.
 - It learns the relationship between users and movies by minimizing reconstruction error.
5. **Movie Recommendation**
 - After training, the RBM predicts movies that a user is likely to enjoy but has not yet watched.
 - These predicted movies are recommended to the user based on learned preferences.

Advantages of Using RBMs

- Does not require labeled training data.
- Learns hidden user preferences automatically.
- Captures complex relationships between users and movies.
- Improves recommendation accuracy through personalized suggestions.
- Can handle large datasets efficiently.

Conclusion

Restricted Boltzmann Machines (RBMs) provide an effective **unsupervised learning** approach for movie recommendation systems. By learning hidden patterns from users' viewing histories, RBMs can recommend movies that match individual preferences

without requiring labeled data, leading to more personalized and accurate recommendations.

5. Deep Learning Applications, Representation Learning

A manufacturing company wants to detect defective products from sensor readings collected every second. Design a deep learning solution using concepts from representation learning and neural networks. **Answer: Deep Learning Applications and Representation Learning**

A manufacturing company collects sensor readings every second to monitor product quality. Detecting defective products manually is difficult because of the large amount of data. A **deep learning model** using **representation learning** can automatically learn patterns from sensor data and accurately identify defective products.

Deep Learning Solution

1. **Collect Sensor Data**
 - Gather sensor readings such as temperature, pressure, vibration, and speed from the manufacturing process.
 - Preprocess the data by removing noise and normalizing values.
2. **Representation Learning**
 - Use a deep neural network to automatically learn important features from the sensor readings.
 - The model extracts meaningful patterns without manual feature engineering.
3. **Neural Network Model**
 - Train a **Deep Neural Network (DNN)** or **Autoencoder** using historical sensor data.
 - The hidden layers learn the difference between normal and defective product patterns.
4. **Defect Detection**
 - During testing, the trained model analyzes new sensor readings.
 - If the learned representation differs significantly from normal patterns, the product is classified as **defective**; otherwise, it is classified as **normal**.
5. **Continuous Monitoring**
 - The model processes sensor data in real time and immediately alerts operators when a defect is detected.

Advantages

- Automatically learns important features from sensor data.
- Detects defects with high accuracy.
- Reduces manual inspection and human error.
- Supports real-time quality monitoring.
- Improves production efficiency and reduces manufacturing costs.

Conclusion

By using **representation learning** and a **deep neural network**, the manufacturing company can automatically extract useful features from sensor readings and accurately detect defective products. This approach enables real-time monitoring, improves product quality, and reduces production losses.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand